ELIXIR's role in the cancer landscape

ELIXIR All Hands , 5-8 June 2023, Dublin, Ireland

Over 2.7 million people are diagnosed with Cancer every year in the European Union, with 1.3 million dying due to this annually. Cancer cases are predicted to rise by 24% by 2035 due to our ageing populations and lifestyles. In an effort to mitigate this the EU has launched its Mission on Cancer and Europe's Beating Cancer Plan. ELIXIR is contributing to these by working to develop infrastructure to support researchers with accessing and analysing cancer data, which will improve the value and impact of life science research on public health and the economy.

Development of standards, services, tools and workflows

canSERV
providing cutting edge cancer research services across europe

Aiming to streamline cancer research and provide free access to a comprehensive set of oncology research services provided by European Research Infrastructures.

Researchers will be able to select from a bespoke catalogue of over 400 services that include access to disease models, clinical trials, biomarker research, training and more.

ELIXIR leads WP9, which will harmonise the provided digital services with best practices in FAIR data and software reuse, and making canSERV research outputs reusable by the cancer community.

Beacon v2 enables enhanced querying for cancer datasets across the Beacon network.

Deployment of infrastructure: building capacity

The GDI project will enable at least 15 countries to deploy technically operational 1+MG infrastructure by 2026

The B1MG project has aided cancer proof of concept workflow of services and standards

EUCAIM will build an atlas of FAIR, de-identified real world cancer images from across 15 countries

With these projects we are aiming for interoperability between infrastructures from conception.

A key component will be aligning with the European Health Data Space regulation to enable secondary use and reuse of healthcare data, and to drive interoperability between genomic and health data spaces.

Demonstration: user journey using ELIXIR infrastructure

eosc | cancer

Aims to make a platform enabling storage, sharing, access and analysis of research and clinical cancer data

It will interconnect datasets relevant to cancer research and facilitate adoption of FAIR principles across data sources and research software.

Digital tools, data analytics and AI/ML methods will be integrated with the cancer analysis environments.

These data and analysis systems will be made easily accessible to basic and clinical scientists in open cancer analysis portals.

A long-term service roadmap will be developed, which will set out operating models, responsibilities and access mechanisms for the services developed in this project. The roadmap will be co-developed together with key stakeholders, including EOSC, the 4.UNCAN.eu CSA project, and EHDS pilot actions

ELIXIR Cancer Data Focus Group

Meets monthly on the 4th Thursday, 3pm BST

Aims:
- Highlight the opportunities arising from data, tools and platform interoperability
- Promote the added value from unobstructed data sharing of cancer data for biomedical research and applications.
- Present "Gold Standard" examples from ELIXIR Nodes working with cancer related data

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